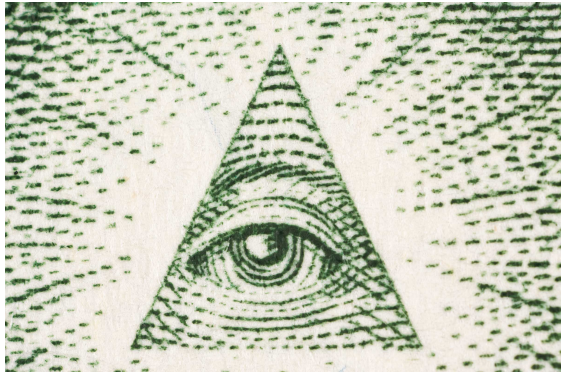
1.1 Computer Concepts

How to use computers, mainly for Windows/Linux but still important for Cisco

by Clement Chan, typed by Nathan Chen

The I: Drive



What the heck?

- Troy stores it’s cyber related things in what we refer to as the I drive
  - Mapped to I:/ for cyber accounts
  - There is a manual path
- What is in the I drive?
  - Blank Images
  - Practice Images



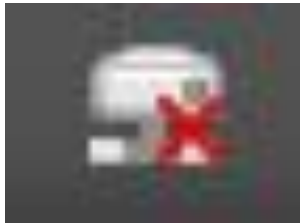
I drive Specifics

Etiquette

- Do not put your work on the I drive
- Do not extract to the I drive
- Do not delete things on the I drive
    - (unless you put them there by accident)

Troubleshooting

- The drive is a network drive, make sure you are connected
- Make sure you are logged into the right account
- You can type file path if not mapped



The Concept of Compression



What the heck are these?

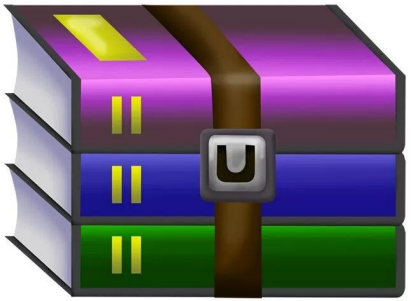


← "7 Zed"



← your basic ZIP

win rawr →



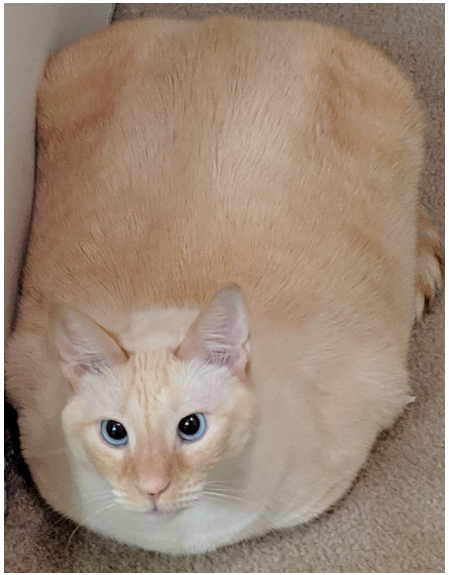
What is file compression?

- Make files smaller by removing redundant info
- Compression also removes “unnecessary data”
- ie: AAAAAABBCCC can be compressed into A6B2C3

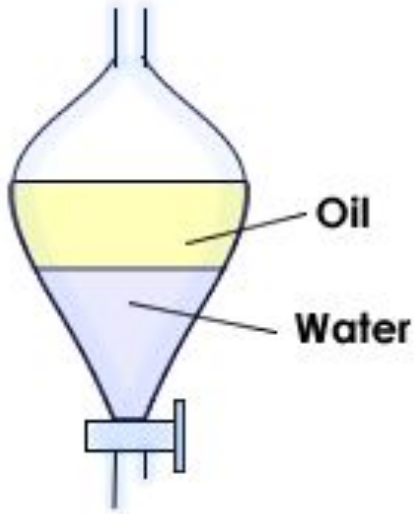


wHy dOEs tHiS reLaTe tO cYbeRpAtriOt?

- Cyberpatriot competes on VM’s
  - VM’s are usually chonky →
  - This only applies for Windows and Linux people
    - Windows is chonkier than Linux
- If you can’t extract the file, you can’t compete!!1!!1



The Extraction Process



HEY YOU

YEAH YOU



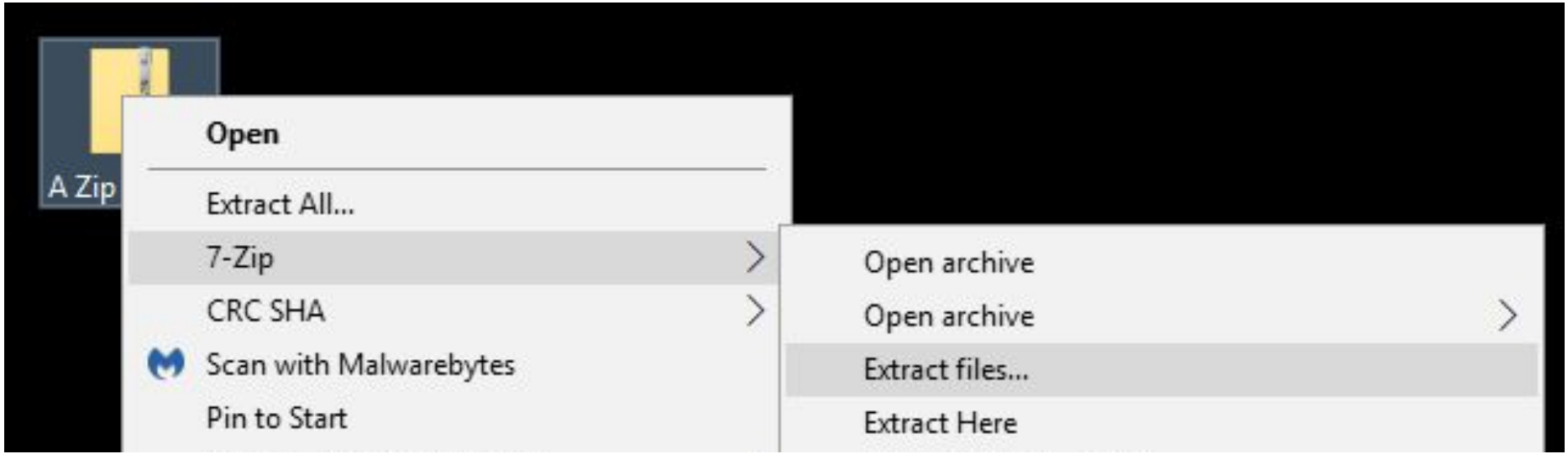
- Do you know how to extract a competition image?
  - You should know that most people last year didn’t so PAY ATTENTION
- THIS WILL BE ON ACTUAL COMPETITION
- I AM TYPING IN CAPS

Step By Step: Here is our file



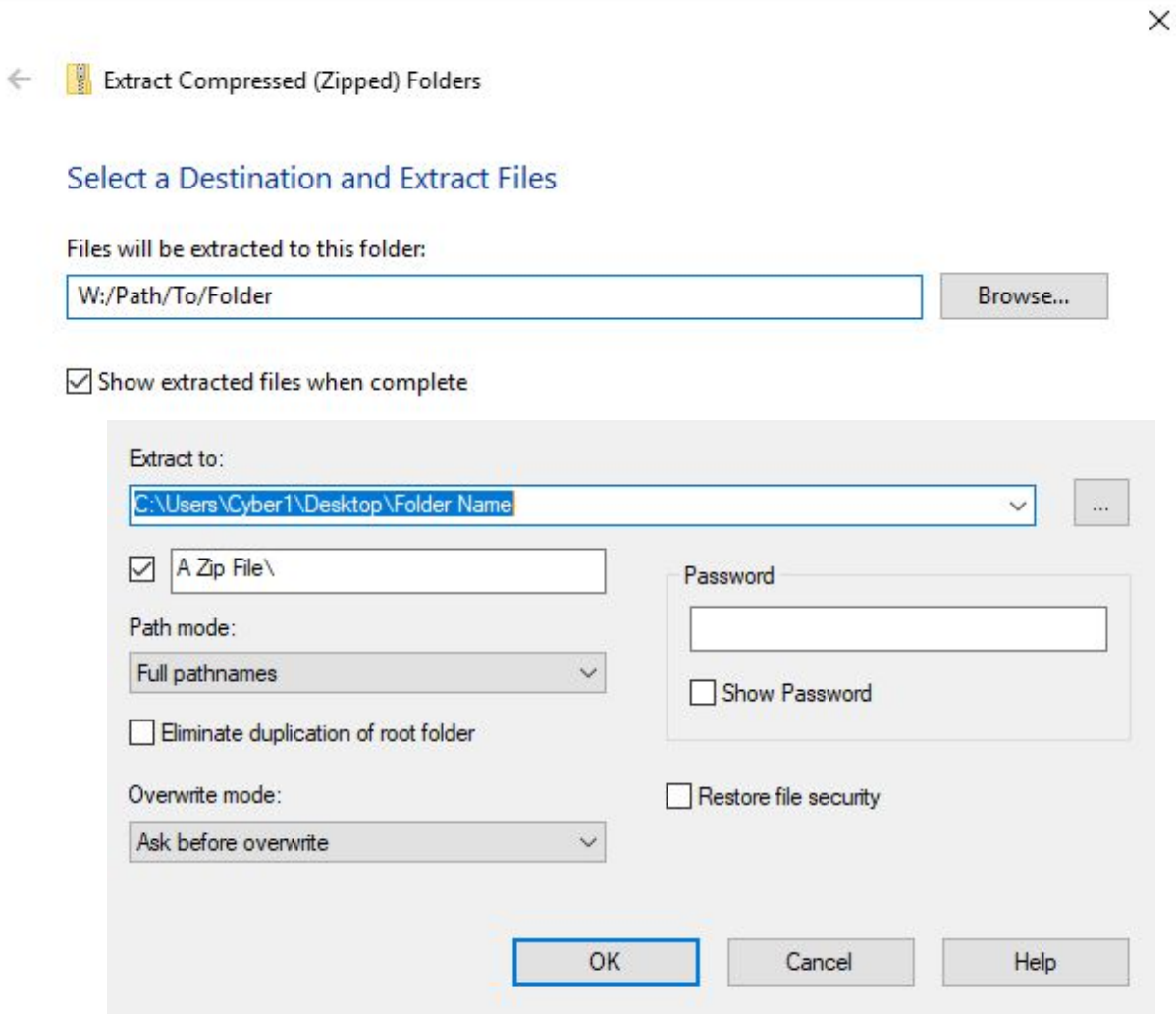
Right Click, 7Zip, Extract files...

If it is only a zip file you can click Extract all (Windows 7 and up)



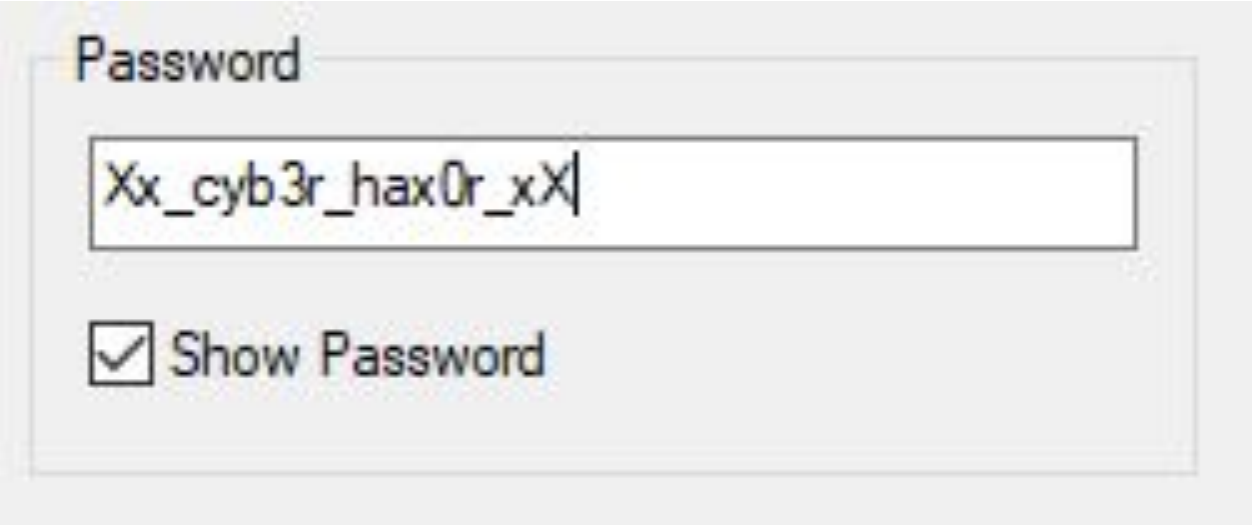
Select Location

- NOT ON THE I DRIVE
- Location depends on what room you are in
- 1001/1002: Desktop is okay
- 403/425: W Drive, not C drive



Password

Cyberpatriot VM’s have a password so enter that in before continuing

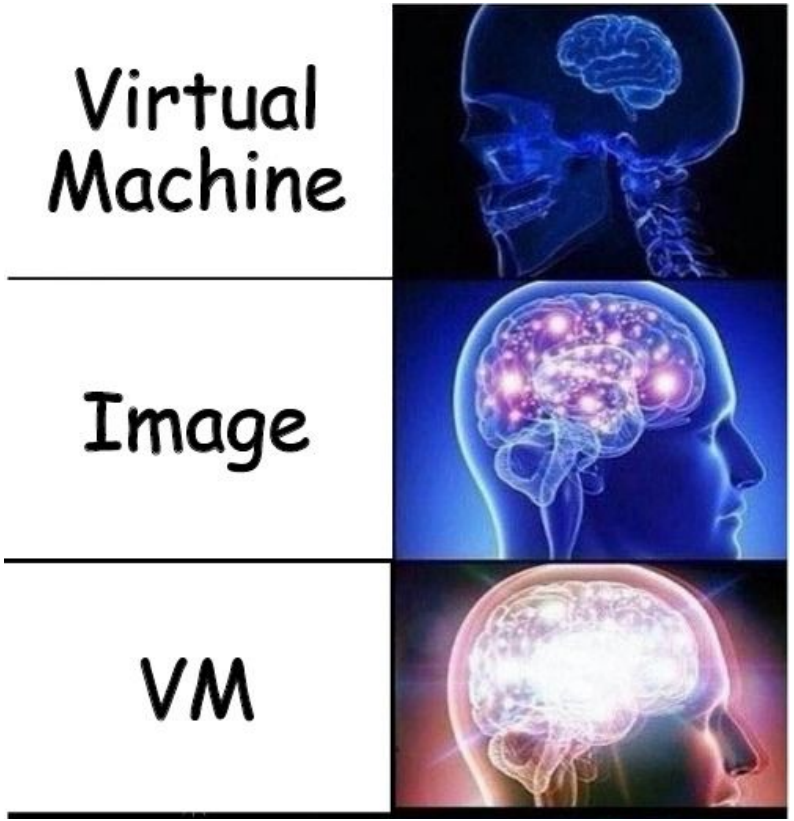


Approve by clicking OK to confirm that you are okay with extracting the files

Virtual Machines

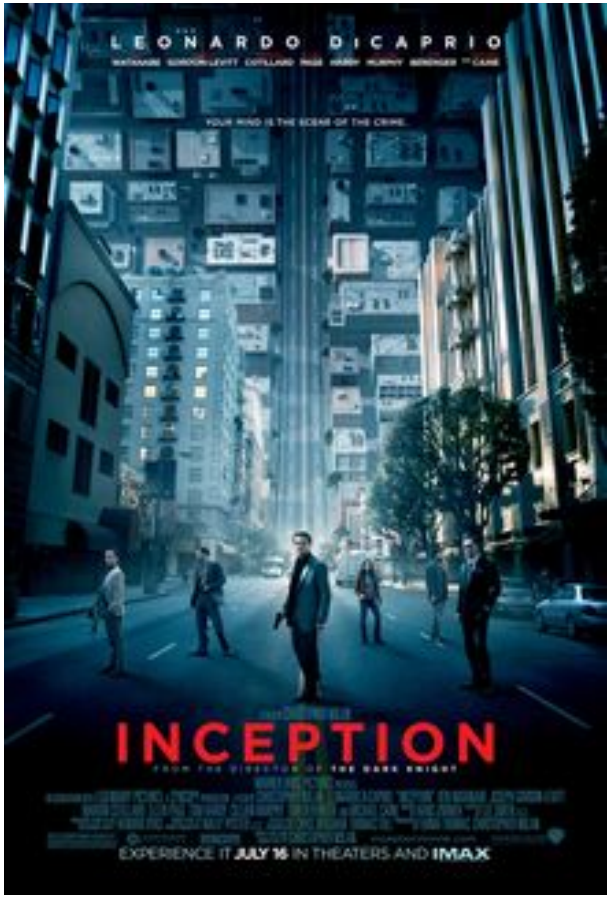
Machines that are virtual





Think of a computer OS inside of a computer OS

A virtual machine is an emulator for operating systems like Windows and Linux



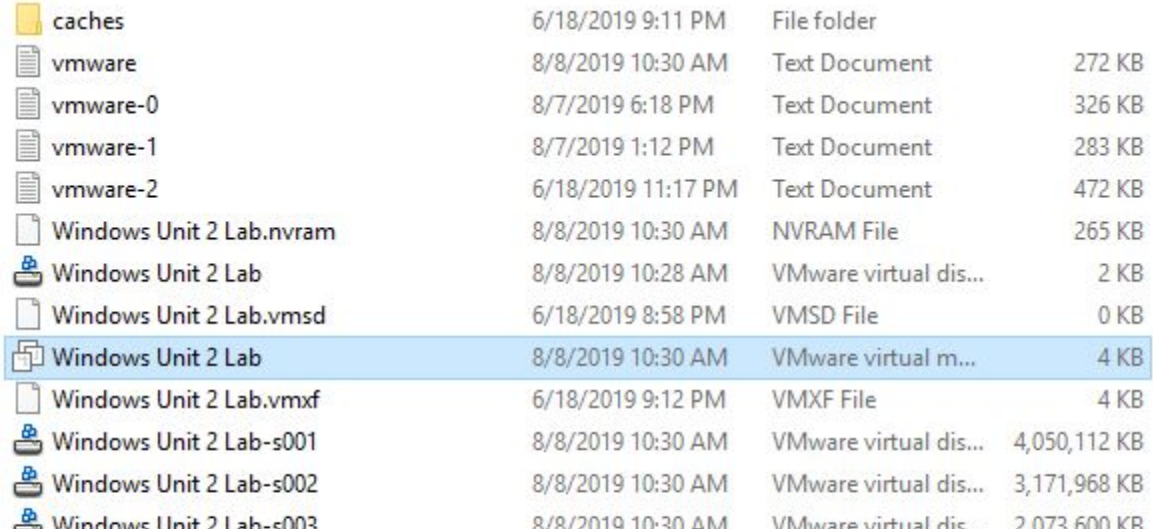
VMWare

What is this virtual machine ware you speak of?

- Cyberpatriot uses free software called VMWare player to host images
- You can download the version used in competitions at home through Cyberpatriot’s website

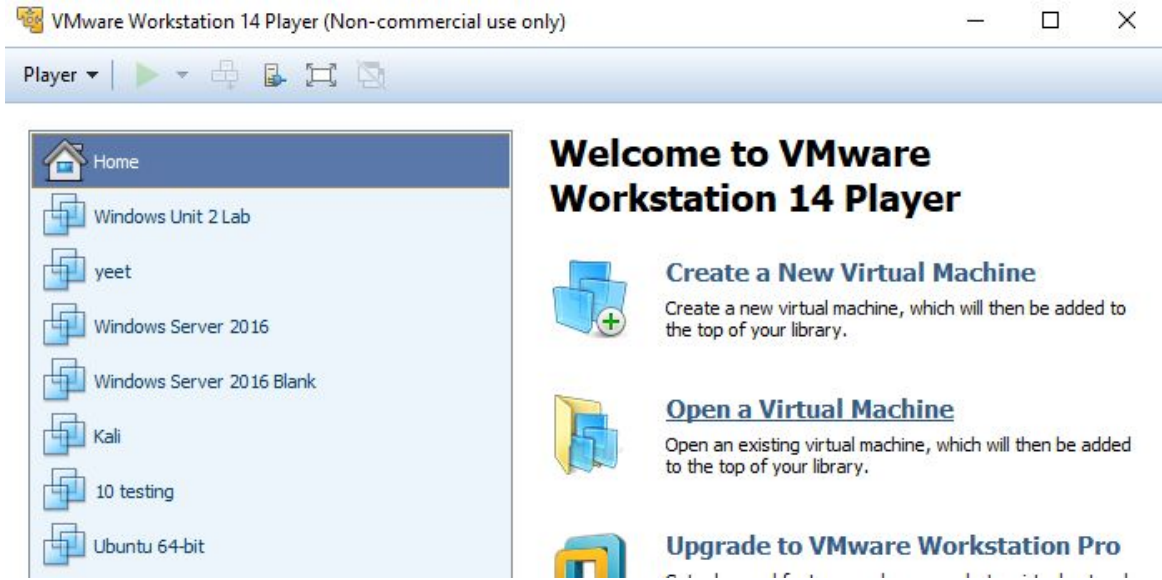
Opening a VM - Method 1 (Lazy)

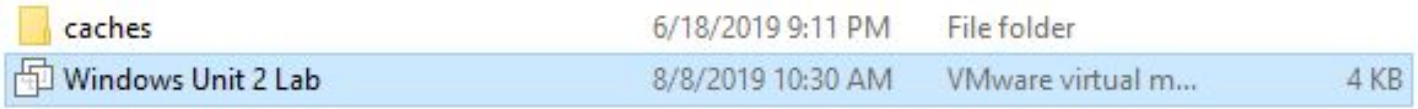
- Go to the location of your extracted image (you should not see “Compressed size” in File Explorer)
- Find the triple square icon (the .vmx file)
- Double click on it to open



Opening a VM - Method 2 (for true h4ck0rz)

- Open VMWare Workstation 14 Player
- Click on “Open a Virtual Machine”
- Browse to the extracted image folder
- Find the triple square icon
- Double-click to open





Advantage of Method 2

Method 2 won’t immediately open your virtual image - it’ll just load a virtual image profile for you to open from.

This way you can edit VM settings like the amount of RAM and processor cores to use. You can boost the performance of the virtual machine.

CyberPatriot starts the timer once you open the image. It’s a good idea to configure the machine so it performs faster first, then start the image.

VM settings

You cannot change the amount of RAM or processor cores you have while using the VM.

CyberPatriot configures their images to a minimum performance level. Our school computers have either 8GB or 16GB of RAM, so you can use 4GB or 8GB of RAM and 2 processor cores depending on your computer.

You can’t have more RAM than what your computer has. Make sure to leave some RAM for your host, else your computer will die

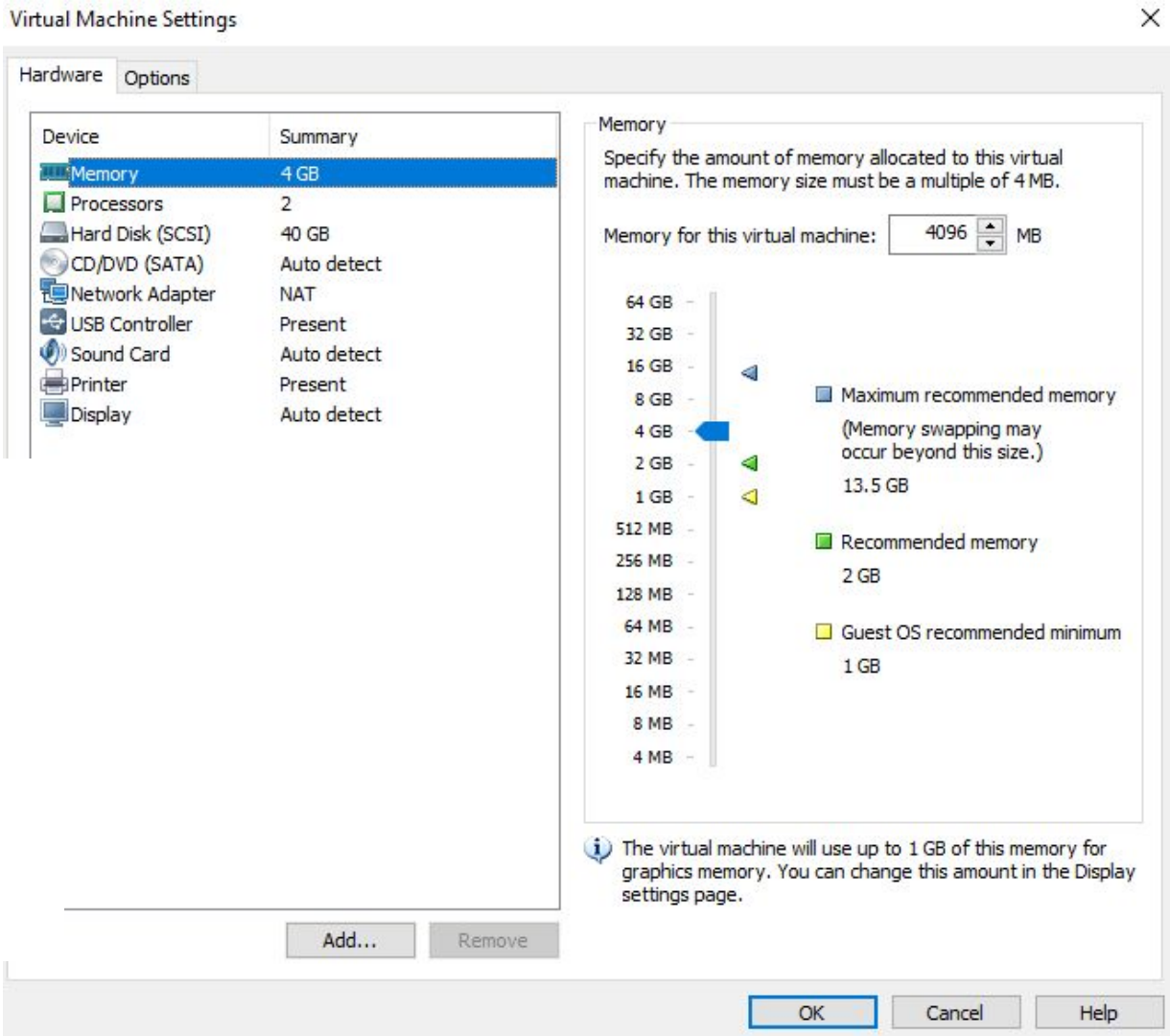
Editing VM settings

Windows Unit 2 Lab

State: Powered Off
OS: Windows 10 x64
Version: Workstation 14.x virtual machine
RAM: 4 GB

Play virtual machine

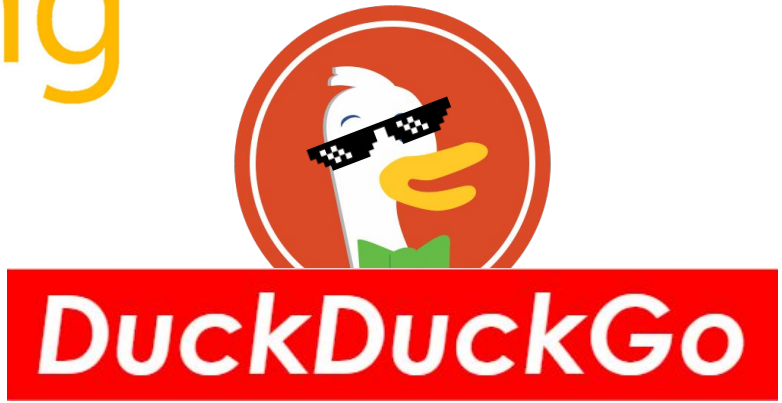
Edit virtual machine settings



Search Engines

Bingbong

Goo.gl



Ok Google, what is the answer to life the universe and everything

CyberPatriot is open note, meaning Google/Bing is ok to use

LMGTFY: Let me Google that for you

If you don’t know anything about something, Google it

But there is one problem, the school’s network has a proxy that messes up Google’s certificates inside virtual machines, which is why we either use DuckDuckGo or AskJeeves



How to Google



Use the internet for research, but remember not everything on it works/is correct

Self-research is important for Cyberpatriot.

Some things to note

- Sharing online
  - You cannot take other people's work regarding CyberPatriot
  - You can take people's work that is not CyberPatriot related if they let you
- Looking up stuff
  - I'm stuff
  - cyberpatriot! omg! no!
  - *laughing* troy cyber ur gf is awesome

Fin.

thx